

Research Article

Detecting Deepfakes Using Two-Stream Framework Based on EfficientNet and Vision Transformer

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The ever-growing threat of deepfake technology has stimulated research in deepfake forensics to identify the genuineness of media materials on the Internet. The majority of existing techniques employ the convolutional neural network (CNN) framework to identify deepfake by capturing the image's local features. However, due to the limited scope of receptive fields, relying solely on the feature extraction capabilities of CNNs is insufficient to meet the detection demands posed by increasingly diverse forged samples in real-world scenarios. To address these challenges, this study introduces an innovative model for deepfake identification, leveraging the strengths of EfficientNet and vision transformers to integrate local and global features. This integration allows for more comprehensive learning of deep image features, thereby enhancing detection accuracy. To further enhance detection capabilities and bolster the resilience of our approach against image compression, we designed a dual-stream architecture that incorporates noise features extracted from steganalysis rich model filters as an additional input. By integrating multiple information streams, the model's performance is effectively improved. From the experiments, the proposed model achieved high accuracy rates of 0.845, 0.999, and 0.815 on the video test sets FaceForensics, Celeb-DF-V2, and Wild Data, respectively. It also attained the best results on the photo test sets FaceApp, FaceSwapApp, and FaceFuse, with scores of 0.998, 0.925, and 0.918. Notably, the model demonstrated strong performance across various deepfake datasets, especially those sourced from social networks. With 88M parameters compared to 101M in the next-best competing method, the proposed model confirms its effectiveness in real-world case studies while maintaining reduced computational complexity.

1. Introduction

“Deepfakes”: Seeing is not always believing. The expression “deepfake” blends “deep learning” with “fake.” It was first introduced by a Reddit user known as “deepfakes,” who gained notoriety in December 2017 for posting manufactured videos of Scarlett Johansson and other actresses on the platform [1]. This event marked the rise of deepfake, and the user's username became synonymous with this technology.

Due to the universality of deepfakes and the high degree of open source, the creation of fake videos necessitates merely a handful of facial photographs to accomplish a swap

of faces in the footage [2–5]. Although deepfake has its positive uses, it is unavoidable that certain ill-intentioned individuals will gather information accessible online and exploit it for malevolent objectives. For example, influential people such as actresses, politicians, and company executives are faked with corresponding negative videos and disseminated on the Internet to mislead public opinion, win votes, manipulate stock prices, etc. [6, 7]. In addition, with the maturity of deepfake technology, quick face-swapping apps such as ZAO [8] and FaceApp [9] have appeared on the market one after another. In June 2019, a one-click stripping software program named DeepNude emerged, bringing serious negative impacts to the society.

The aforementioned problems emphasize how crucial it is to be able to detect deepfakes. As a result, research in the field of deepfake detection is being invigorated to verify media content's reliability and genuineness. Scholars have successively worked on constructing datasets for deepfake identification, including FaceForensics [10], Celeb-DF [11], DeeperForensics [12], and so on, to carry out research studies for deepfake detection. Moreover, Facebook, together with Microsoft and Amazon, held the global deepfake detection competition (DFDC) [13], which led to the emergence of different innovative mechanisms to detect deepfakes. However, despite the advantages of each proposed mechanism, current detection methods still have various challenges, and the majority fail to effectively handle the latest deepfake algorithms. Furthermore, most visual content found online is in compressed form, which further makes deepfake detection more difficult. Taking this into account, forensics on compressed deepfakes becomes meaningful and challenging work.

Given that the work in [14–18] demonstrated the outstanding effectiveness of dual-stream model structures across multiple fields and emphasized the importance of dual-stream networks in handling complex tasks, our research introduces a dual-stream framework specifically designed for deepfake detection. It capitalizes on the unique property that image noise information remains resilient when typical detectors experience a steep drop in performance due to loss of feature information from image compression. The framework takes advantage of the vision transformer in learning the global pixel-to-pixel relationship, integrating pixel spatial interconnections on a global scale with localized visual attributes in an affordable approach to identify deepfake. The key contributions of our study presented within this article can be encapsulated as follows:

1. We introduce a dual-stream network structure for deepfake forensics to learn hidden traces from different types of image information in terms of local features and global relationships of pixels.
2. We adopt noise feature maps as additional inputs to improve deepfake detection performance.
3. Our method has been verified across multiple datasets, and the outcomes of these experiments illustrate that our proposed approach yields proficient results on diverse benchmarks, with particularly standout performance on Wild Data where it attains peak efficacy.

In summary, we explored EfficientNet and vision transformer to design a two-stream deepfake detection model that has an excellent performance in our experiments. We consider both modalities, RGB and noise features, in the input phase, and the outcomes suggest that our framework delivers superior results, even in the absence of preliminary training.

The organization of this article is presented as follows. Section 2 provides a brief overview of recent developments in deepfake creation and identification techniques. Section 3 delves into the structure of our suggested deepfake detection model, along with its detailed deployment procedures. Section 4 is a summary of the overall experimental design

and an evaluation of the results. Lastly, Section 5 furnishes an extensive recapitulation of the paper in its entirety.

2. Related Works

In the past few years, neural networks for deepfake technology have drawn substantial interest. In particular, the generative adversarial network (GAN), with its generator and discriminator structure, makes deepfake more realistic and deceptive. Noteworthy methods include PGGAN [19], StarGAN v2 [20], FSGAN [21], StyleGAN [22], CycleGAN [23], and RecycleGAN [24]. These advancements enable deepfakes to achieve striking authenticity in both static images and video sequences. While this technological evolution amplifies societal and ethical risks, it has simultaneously spurred rapid advancements in adversarial detection techniques. Current research on deepfake identification has coalesced into two primary technical paradigms: convolutional neural network (CNN)-based methods for localized feature analysis and transformer-based approaches for global semantic modeling, with significant progress markedly enhancing the robustness of detection systems.

2.1. CNN-Based Detection. Owing to CNN's remarkable capability in extracting local features, it is extensively applied for image forensic tasks [25]. For example, Afchar et al. [26] presented a novel approach called MesoNet for identifying deepfakes, which constructed a compact CNN model using shallow convolutional networks from image mesoscopic level features. Chollet [27] introduced the XceptionNet framework alongside the launch of FaceForensics data, demonstrating unparalleled efficacy in the dataset's comparative evaluations. In [28], ResNet50, ResNet101, ResNet152, and VGG16 have been applied to detect deepfake, stemming from the mismatch between the facial area and the adjacent context. Similarly, Nguyen et al. [29] introduced a Capsule-Forensics architecture, reliant on VGG, intended to identify various forms of deepfake. Additionally, Qian et al. [30] introduced frequency as the complementary viewpoint for deepfake detection and proposed the Frequency in Face Forgery Network (F^3 -Net). Deng et al. [31] introduced a new network by improving and optimizing EfficientNet-V2 to detect deepfake and have achieved high prediction accuracy. Li et al. [32] employed face X-ray as a new image representation to display the deepfake synthetic boundary. Other scholars have also considered the use of attentional mechanisms for deep forgery detection. For instance, Khormali and Yuan [33] introduced a deepfake detection model that utilizes attention mechanisms and is comprised of two distinct modules, extracting more discriminative information from exploits of the fine-grained and spatial locality attributes. Dang et al. [34] introduced a new attention mechanism known for its significant adaptability applied to the baseline networks to make deepfake detection results look better. Though the majority of deepfake detection methods are based on CNNs, scant attention has been directed toward delving into more complex technologies such as the vision transformer.

2.2. Vision Transformer. The exceptional effectiveness of the transformer model within the domain of NLP has inspired scholars to explore its application in computer vision tasks. For example, Khan and Dai [35] designed an approach with incremental learning using XceptionNet [27] and vision transformer for deepfake detection. Wodajo and Atnafu [36] proposed a convolutional vision transformer in which vision transformer uses the learnable features extracted by CNN for processing and categorizes them using an attention-based system. To obtain better detection performance, Wang et al. [37] employed vision transformer as the core network infrastructure and constructed a module for cross-modality fusion that combines frequency details with RGB characteristics for deepfake detection. In addition, Heo et al. [38] suggested a fresh strategy for identifying deepfakes by adding the distillation of EfficientNet features on the basis of vision transformer. Furthermore, Khormali and Yuan [39] have taken an effort to construct a comprehensive deepfake identification system to learn hidden traces from information at different scales. Although these methods are highly effective, there is still room for further investigation into transformer-based approaches within the realm of deepfake detection. Consequently, this study designs a dual-stream model based on the transformer technology.

3. Proposed Model

This section furnishes a comprehensive explanation of the devised deepfake identification methodology built upon multiple feature inputs. The suggested approach is designed by applying the transformer model to discern the distinctive attributes of the abnormal traces from the global relationships of pixels under the local image features with RGB input and noise input. Figure 1 visually represents the comprehensive structure of our proposed method.

3.1. Introduction of Noise Information. Deepfakes are generated in a variety of ways, and just analyzing RGB channel information for detection is not sufficient for all situations, so we can consider using other images' information to detect deepfake artifacts. On the one hand, it has been shown that the generative model of the deepfake algorithm will leave specific noise when generating forged images, which inspires us that noise information is a good choice. On the other hand, real-life deepfakes are usually compressed to different levels, which brings difficulties in detection, but this adverse impact can be mitigated by employing the noise characteristic present in the image [40, 41]. However, limiting the analysis for deepfake detection exclusively to the image's noise feature map results in a loss of most local information, which is detrimental to the classification learning task.

Therefore, in order to design a detection network with both generalization and robustness, this paper adopts a two-stream architecture that combines RGB information and its noise information extracted by SRM filters, so as to learn image deepfake artifacts from different dimensions.

SRM, as a steganalysis technique, is based on the spatial rich model and employs multiple submodels to extract more types of features. Reference [40] finds that utilizing just 3 kernels yields satisfactory results, while employing all 30 kernels does not provide substantial performance enhancement. Thus, we adopt the pattern of 3 kernels to obtain the image's noise feature map, which then feeds into the noise stream.

3.2. Parallel Structure

3.2.1. Preprocessing. In a general classification task of image processing, while forgeries may appear dispersed throughout various regions of the image, deepfake approaches primarily focus on altering facial regions, leaving the background regions unchanged. Therefore, focusing the training of the deepfake detection model exclusively on facial regions not only reduces computational demands but also enhances model efficacy because of the elimination of the background area's influence [10]. Initially, 30 frames are chosen from each video at regular intervals, one after the other, to minimize redundant frames while increasing data diversity, and next, RetinaFace [42] is employed to obtain facial landmarks, an advanced facial detection method. In the end, each frame's facial area is cropped, resized, and adjusted using the determined landmarks [43].

3.2.2. Feature Extraction and Fusion. Taking into account the performance of ResNet-50 and MobileNetV3-Large in terms of accuracy and efficiency, EfficientNet-B0 (the most compact model within the EfficientNet series) achieves a better balance with higher accuracy, fewer parameters, and lower computational cost. Therefore, we employ it as the convolutional feature extractor to handle the input ($X \in \mathbb{R}^{HW \times 3}$) and obtain important low-level local information. Though expressing the EfficientNet-B0 in a formula, we can regard its convolutional layer as the following function mapping:

$$Y_i = F(X_i). \quad (1)$$

To fully utilize the local features obtained from the RGB input ($Y_{\text{RGB}} \in \mathbb{R}^{HW \times C}$) and the noise input ($Y_{\text{SRM}} \in \mathbb{R}^{HW \times C}$), the idea is to fuse them. Given that images must be transformed into a sequence of uniform-sized patch embeddings before analyzing them with the transformer model, instead of directly fusing Y_{RGB} and Y_{SRM} , we take a cue from transform [44] and convert the two features into fixed-size patches (16×16 patch size is used) first by the linear project. Each patch is then flattened to a 768-dimensional vector, which helps in standardizing the feature representation for subsequent processing. These patches are subjected to a Concat operation, and since the obtained patches still need to follow the order of the patches, we adopt learnable positional encoding to get each token, denoted as $Z \in \mathbb{R}^{M \times d}$ where M and d are the number and dimension of patches, respectively. In addition, following the aforementioned operations to fully use the available information for image classification, we designed an Efficient → Former

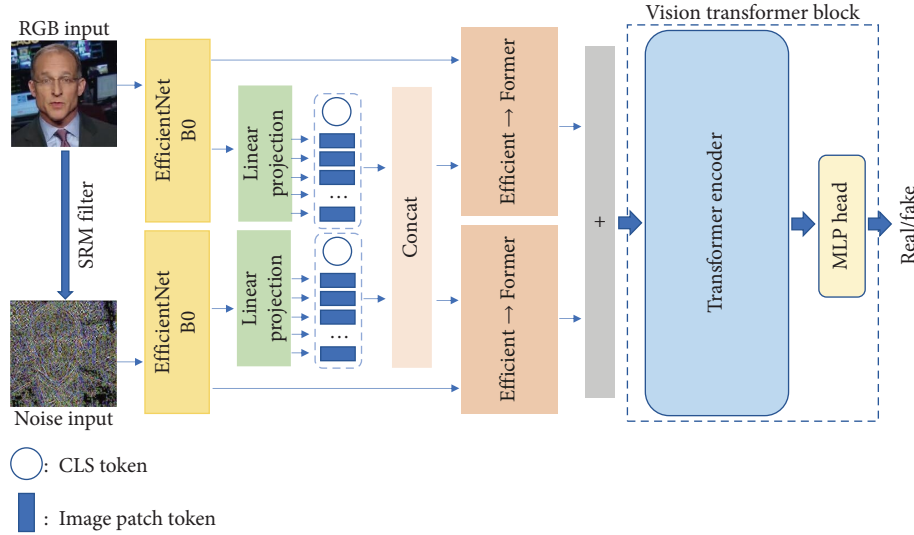


FIGURE 1: The architecture of the proposed method. The noise input is generated by processing the RGB image through a steganalysis rich model (SRM) filter, resulting in a noise feature map. These noise features, along with the original RGB input, are processed together using EfficientNet and then serve as inputs to the vision transformer (ViT). This integration provides complementary evidence, enhancing the effectiveness of the model for deepfake detection.

module. Its output is superimposed with the earlier tokens, serving as the input for the final transformer encoder block.

3.2.3. Efficient-Former. Referring to an effective way of fusing local and global features proposed in [45], our Efficient-Former block is designed with two inputs. One is the local feature map $X \in \mathbb{R}^{H \times W \times C}$ which has C channels over height H and width W , and the other is the tokens $Z \in \mathbb{R}^{M \times d}$ mentioned in Section 3.2.2. The Efficient-Former outputs the global tokens Z' , and the structure of the Efficient-Former block is shown in Figure 2.

Specifically, the computational procedure of this structure follows the standard attention function [38]:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (2)$$

where Q , K , and V represent query, key, and value, respectively. Similarly, Z' is computed as

$$Z' = \text{Attention}(ZW^Q, X, X)W^O, \quad (3)$$

where W^Q represents the query matrix and W^O is used for data merging. As demonstrated in [43], such a design choice (where K and V share the same source) effectively captures spatial dependencies in image-based tasks without introducing redundant parameters. This approach has been widely adopted in vision-specific architectures to balance computational efficiency and representational power.

4. Experiments and Results

Most of the existing deepfake detection methods predominantly undergo training and assessment on a singular public deepfake dataset to ascertain their detection

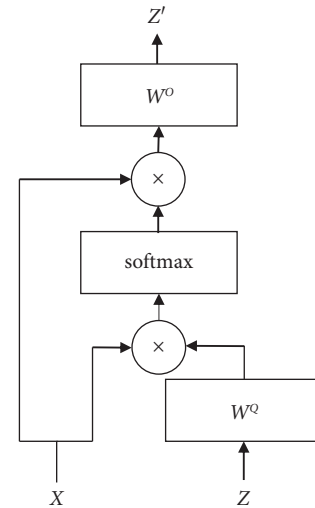


FIGURE 2: Efficient-Former block. A lightweight feature fusion method.

proficiency. Although these approaches showed high detection accuracy in their experiments, it does not guarantee that they will perform similarly when dealing with various other deepfake datasets. In addition, adversarial actors are continually developing novel methods to produce increasingly convincing deepfakes, which poses a greater challenge for deepfake detection. Therefore, taking the above problems into account, we collected the public deepfake datasets from the Internet and added our private fake face datasets to them for experiments.

4.1. Dataset Construction. Our experimental data include the mainstream public deepfake datasets.



FIGURE 3: Examples of public datasets. (a)–(h) are samples of DeepFakeForensics-1.0 [12], FMFCC-V [48], DFDC [13], FaceForensics [10], Celeb-DF-V2 [11], WildDeepfake [49], Deepfake-TIMIT [47], and DFD [46], respectively.

4.1.1. Publicly Available Datasets

4.1.1.1. DFDC. It is a formal dataset for a contest hosted by Facebook at Kaggle, with a total of 119,196 videos, the original videos were shot with actors, the ratio of real to faked videos was about 1:5, and the videos, spanning approximately 10 s, encompassed a diverse array of scenes [13].

4.1.1.2. FaceForensics. The real videos consist of 1000 videos from YouTube and the corresponding deepfake videos are generated on the basis of the real videos employing four distinct facial modification methods: Deepfakes [50], Face2Face [51], FaceSwap [52], and Neural Textures [53]. Moreover, the dataset includes three tiers of video quality, each subjected to varying degrees of compression: uncompressed videos (C0), mildly compressed videos (C23), and videos of inferior quality (C40).

4.1.1.3. Celeb-DF-V2. The Celeb-DF-V2 dataset contains 5639 high-quality deepfake videos generated from 590 YouTube real videos using the improved deepfake synthesis algorithm.

4.1.1.4. DeepFakeForensics-1.0. A comprehensive deepfake dataset featuring 50,000 original videos alongside 10,000 altered videos was released by Nanyang Technological University and Shangyang Technology.

The other public datasets such as Deep Fake Detection (DFD) [46], Deepfake-TIMIT [47], FMFCC-V [48], and WildDeepfake [49] are also adopted. Figure 3 shows some of their samples. For convenience, their basic information is presented in Table 1.

4.1.2. Private Datasets. The private datasets used for experiments were carried out considering images/videos created by DeepFaceLab [4], FSGAN [21], StyleGAN [22], and other deepfake algorithms in conjunction with other deepfake applications: ZAO [8], FaceApp [9], FaceSwap [52], etc. for deepfake of faces.

4.2. Training and Testing Details

4.2.1. Training Details. For the dataset mentioned in Section 4.1, given that the quantity of authentic and fabricated videos within each dataset is unequal, we choose videos from the dataset in different proportions to ensure a balanced distribution of data volume. Also, to enhance the variety and multiplicity of training dataset, for all the selected real and forgery videos, we extract a different frame of the video at equal intervals when the video is segmented into individual frames. Furthermore, to allow the network to comprehensively grasp the facial features and adjacent areas in both authentic and manipulated images, we use one-stage inference face detector (RetinaFace) [42] to crop the face regions as inputs with a size of 380×380 , resulting in 1,820,114 face images, as our training set.

TABLE 1: Information of the public datasets.

	Real	Fake
DFDC [13]	19,196	100,000
FaceForensics [10]	1000	4000
Celeb-DF-V2 [11]	590	5639
DeeperForensics-1.0 [12]	50,000	10,000
DFD [46]	363	3068
Deepfake-TIMIT [47]	320	640
FMFCC-V [48]	44,290	38,102
WildDeepfake [49]	3825	3509

TABLE 2: Evaluating the efficacy of our suggested approach through frame-level examination by comparing its performance with that of other deepfake detection approaches on various datasets.

Methods	FaceForensics [10] ACC/AUC	Celeb-DF-V2 [11] ACC/AUC	Wild Data ACC/AUC	# params (M)
Davide [54]	0.902/ 0.926	1.000 /0.997	0.812/0.963	101
Davide' [54]	0.919 /0.905	0.998/0.984	0.802/0.926	
EfficientNet-B7 [55]	0.816/0.868	0.999/0.995	0.787/0.922	64
EfficientNet-B7' [55]	0.716/0.816	0.994/0.877	0.750/0.862	
Proposed Method	0.845/0.917	0.999/ 0.998	0.815/0.964	88

Note: Utilizing a model that has undergone prior training is indicated as Davide [54], the absence of a pretrained model is denoted as Davide' [54], and the same is indicated for EfficientNet-B7 [55] and EfficientNet-B7' [55]. Wild Data is the dataset we downloaded from the Internet, containing deepfake videos of various qualities, characters, and scenes. All datasets used in the experiment were kept identity-disjoint, ensuring no subject overlap across datasets. The bold values indicate the best performance for each metric.

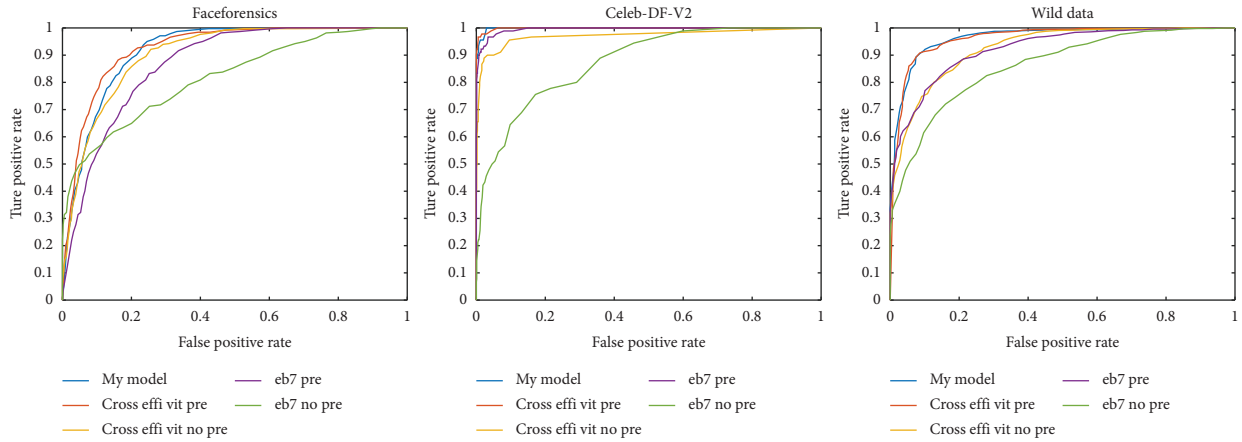


FIGURE 4: Receiver operating characteristic (ROC) plots for the network models of three different types of deepfake datasets.

Data augmentation techniques are applied, including image compression, adding Gaussian noise, Gaussian filtering, random flipping, random padding, grayscale conversion, and random affine transformation, to enrich the training samples.

All tests were carried out on a computer configured with four Quadro RTX 8000s. The training procedure of our proposed method employs the SGD optimizer with a weight decay of $1e-7$, combined with the StepLR scheduler that reduces the learning rate by a factor of 0.1 ($\gamma = 0.1$) every 15 epochs. The initial learning rate is set to 0.01, with a batch size of 128 and a total of 22 training epochs.

4.2.2. Testing Details. The evaluation set is independent from the training set. Following the identical approach

applied to the training set, we extract 30 frames from each video in the test set. As a result, each video is indicated whether it is real or not by the average of the predicted values of 30 frames. For classification, considering the fairness of the experiments, we set all the true/false thresholds to 0.5.

4.3. Comparison Experiments. To provide better insight into the performance of our proposed deepfake detection method, we compared our model with two other methods, one of which stands out as the top-performing model [54] on the DFDC dataset shown in [58], and the other is an EfficientNet-B7-based method proposed in [55]. It should be highlighted that both of these methods employ characteristics derived from the pretrained model; therefore, to ensure an equitable comparison and to better understand our

TABLE 3: Accuracy of detection models for each type of deepfake image.

Methods	FaceApp [9]	FaceSwapApp [56]	FaceFuse [57]	StyleGAN [22]
Davide [54]	0.793	0.582	0.516	0.909
Davide' [54]	0.519	0.313	0.303	0.767
EfficientNetB7 [55]	0.997	0.910	0.811	1.000
EfficientNetB7' [55]	0.996	0.896	0.901	0.999
Proposed Method	0.998	0.925	0.918	0.999

Note: The bold values indicate the best performance for each metric.

			
Davide [54]	0.864	0.954	0.798
EfficientNetB7 [55]	0.826	0.908	0.709
Proposed method	0.963	0.978	0.914

FIGURE 5: Confidence comparison of detection results for random deepfake samples from the Internet.

proposed method, we reproduced both methods. They were then trained and tested with and without the pretrained features, respectively, using the data in Section 4.2. The optimal outcomes were recorded for comparison with our proposed method.

All experiments conducted in this research encompass two dimensions, including analyses at both the frame and image levels. Performance at the frame level is assessed utilizing two predominant deepfake datasets alongside our own deepfake dataset gathered from the Internet (Wild Data). The results are displayed in Table 2, and Figure 4 illustrates the receiver operating characteristic (ROC) curves for each network model at the frame level per dataset. As indicated in Table 2, on the FaceForensics [10] dataset, despite our proposed method not attaining the top performance, its AUC trails behind the top result from [54] by only 0.009. In addition, we observed that the proposed scheme outperformed the other deepfake detection approaches on both Celeb-DF-V2 and Wild Data datasets. Particularly, our approach stands out for its excellence in both the accuracy and AUC metrics within the Wild Data dataset, and it is marked by a lower complexity in comparison to the model outlined in [54]. Table 3 encapsulates the detection outcomes at the image level. As indicated in Table 3, the proposed approach yields accuracies of 0.998, 0.925, and 0.918 corresponding to FaceApp [9], FaceSwapApp [56], and FaceFuse [57], respectively. Therefore, the methodology outlined in this paper is capable of effectively distinguishing between genuine images and deepfakes.

4.4. Qualitative Results [59]. We randomly selected three deepfake samples from the Internet and evaluated them using the proposed method, analyzing the confidence scores for each image to assess the performance. As shown in Figure 5, in the detection of all three images, the proposed dual-stream framework consistently exhibited high confidence, accurately identifying the manipulated faces. In contrast, the two comparative methods showed significant fluctuations in confidence across the three images, with overall lower and less stable results than the proposed method. Therefore, the proposed method demonstrates significant advantages in enhancing detection accuracy and robustness, providing a more effective solution for practical applications.

5. Conclusion

In this study, we introduce a comprehensive dual-stream model for deepfake detection utilizing both CNN and transformer architectures. The proposed scheme is based on RGB features for image analysis while introducing noise features of images to provide additional evidence for deepfake detection. Then, we have investigated the synergistic benefits derived from combining EfficientNet's prowess in local feature extraction with the transformer's capacity for encoding global information. Experimental results obtained from several benchmarks show that the introduction of noise features as a supporting parameter for deepfake detection is effective, achieving high prediction accuracy on both images and videos. In particular, the peak

performance is attained on the Wild Data dataset, which signifies the adaptability of the structure to a broad spectrum of deepfake detection tasks.

Data Availability Statement

The data are not publicly available because of privacy or ethical restrictions.

Conflicts of Interest

The authors declare no conflicts of interest.

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